



DATA SCIENCE

Weekly Lecture Schedule • Fall 2026

Bachelor of Science in Information Technology • *From Foundations to Intelligent Systems*

Every chapter is seen through four lenses: learning analytics, project management, IoT and cybersecurity

CONTACT HOURS	DURATION	PREREQUISITE	CONTENTS
2 lecture hours + 1 lab hour a week	16 teaching weeks; midterm Week 10, capstone due Week 16	Machine Learning (Fall 2026), taken first	32 chapters in 6 parts; Python labs, and R where it fits

📅	📖 Lecture topics	🏢 Laboratory (1 h)	📋 Assessment
1	1 The Data Science Landscape – what data science is, AI–ML–DL nesting, generative and agentic AI, four levels of analytics, team roles 2 The Data Science Lifecycle – CRISP-DM, framing the problem, two kinds of metric	Meet a dataset; write a frame	—
2	3 Data: Types, Sources and Quality – three kinds of data, measurement scales, tidy data, quality, provenance 4 The Mathematical Toolkit – vectors, functions and gradients, gradient descent	Ten-minute audit; distance and descent	—
3	5 Describing Data – centre, spread, shape, common distributions 6 Probability and Uncertainty – conditional probability, Bayes’ theorem, expected value, central limit theorem	Centre, spread, shape; base-rate simulation	Quiz 1 (Ch 1–4)
4	7 Acquisition, Wrangling and Cleaning – where data comes from, the cleaning sequence, joins, encoding and scaling, data leakage	Leak-proof pipeline	—
5	8 Exploratory Data Analysis and Visualisation – choosing the right chart, bivariate exploration, the third variable, visualisation anti-patterns	Standard EDA pass	Assignment 1 set
6	9 Inferential Statistics – confidence intervals, hypothesis testing, effect size and power, multiple comparisons 10 Relationships and Causality – confounders and colliders, experiments, when you cannot randomise	Intervals, tests, confounder simulation	—
7	11 Machine Learning Fundamentals – rev ML Ch. 1, 3, 7: four paradigms, three-way split, overfitting and the learning curve, baselines 12 Supervised Learning I: Regression – rev ML Ch. 5: error measures, checking assumptions, polynomial fits, ridge and lasso	Split, baseline, diagnose; fit and regularise	Assignment 1 due
8	13 Supervised Learning II: Classification – rev ML Ch. 6, 9–12: from regression to classification, decision boundaries, the threshold as a decision 14 Model Evaluation and Selection – rev ML Ch. 8: confusion matrix, precision against recall, threshold from costs, cross-validation	Five classifiers compared; evaluate honestly with a cost-based threshold	—
9	15 Feature Engineering and Ensembles – rev ML Ch. 14 (features new): feature selection, ensembles, reading feature importances 16 Unsupervised Learning – rev ML Ch. 15–16: k-means, hierarchical and DBSCAN, dimensionality reduction, anomalies	Engineer, ensemble, interpret; cluster and detect	Assignment 2 set
10	Midterm Examination – closed-book; all of Parts I–III (Ch 1–16) finished the week before	Midterm review: baseline to evaluation	Midterm (Ch 1–16)
11	17 Neural Network Foundations – one neuron, non-linearity, layers and the forward pass, the training cycle 18 Deep Architectures – convolutional, recurrent, attention and transformers, autoencoders, transfer learning	Network from scratch; transfer learning	—
12	19 Sequences, Time Series and Streaming – decomposition, forecasting features, temporal validation, stream processing, drift 20 Text and Natural Language Processing – bag of words, TF-IDF, dense embeddings, topic modelling	Decompose and walk-forward; TF-IDF classifier	Assignment 2 due
13	21 Generative AI and Large Language Models – how an LLM works, adapting a model, prompting, retrieval-augmented generation, evaluation 22 Agentic AI – agent loop, memory, multi-agent systems, failure modes	Minimal RAG; agent loop with guardrails	Quiz 2 (Ch 17–22)
14	23 Data Engineering at Scale – pipelines, where data lives, the five V’s, parallelism, data contracts 24 Cloud and Edge Computing – service models, VMs, containers and serverless, cloud–fog–edge, cost	Data contracts; containerise a model	Capstone set
15	25 Data Science for the Internet of Things – IoT architecture, telemetry data, sensor fusion, predictive maintenance 26 Data Science for Cybersecurity – security data sources, alert fatigue, four detection problems, attacking the model	Telemetry to alert; per-entity baselines and evasion test	—
16	27 MLOps – serving patterns, training–serving skew, versioning, monitoring, safe rollout 28 Responsible AI and Data Ethics – where bias enters, fairness metrics, privacy, explainability 29–32 Professional Practice – research methods, project management, communication, capstone	Monitored service; fairness measurement	Capstone due

ASSESSMENT WEIGHTS

20%	30%	50%
Sessional: quizzes, assignments, capstone	Midterm, Week 10	Final-term, all chapters

I Foundations of Data

IV Deep Learning and Modern AI

II From Data to Insight

V Data Science in Systems Context

III Machine Learning Core (Revision)

VI Professional Practice

Chapter numbers refer to the Fall 2026 Data Science handout (Appendix G, Sixteen-Week Teaching Schedule). Chapter 14, Evaluation, is the hinge of the course; Chapters 2, 7, 11, 14 and 28 are relied on by everything downstream. The **rev** badge marks chapters revising *Machine Learning Fall 2026* (a prerequisite taken first); only feature engineering and anomaly detection are new. Midterm covers Chapters 1–16; Quiz 2 covers Chapters 17–22; capstone rubric: Ch.32.